Ch 3 Finite Markov Decision Processes

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I Reward
A. Ultimately you give a reward signal based on the goal
1. Importantly, this reward should not import knowledge on <u>how</u> to do the task (In chess, points come from winning, not taking pieces)
2. If reward is NOT the ultimate goal, the model may achieve small goals and ignore actually winning.
3. These prior knowledge leanings should more aptly be established in a starting policy
B. Reward sources are outside of the agent b/c we do not want it to decree it should get reward similar to an action

II. Returns
A. Formally, G_t is the return, which is simply a function of all earned rewards
1. Simplest is → $G_t = R_{t+1} + R_{t+2} + \dots + R_T$, which works best when states can be individual <u>episodes</u> that end at specific states called <u>terminal states</u> (Eq 3.1)
2. <u>Episodic tasks</u> have definite ends, while <u>continuing tasks</u> have no definite end (games vs robot for example)
B. Continuing tasks cannot use Eq 3.1 b/c $T = \infty$ and reward could be infinite
1. Return is therefore determined w/ a <u>discounting</u> approach

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \quad \text{where } 0 \le \gamma \le 1$$

<u>discount rate</u>

2. Discounting diminishes far away reward for continuous tasks

III. Markov Property
A. A state signal that retains all relevant info, aka everything important to making a decision, presently, is considered to be <u>Markov</u>
B. Markov states theoretically allow complete prediction of future states & rewards from current state knowledge just as well as knowing the complete history

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C. Markov Decision Processes are RL tasks that fully satisfy the Markov property. w/
1. State and action space finite means it is a finite MDP

IV. Value Function

A. Value is how good a certain state or state-action pair is regarding future reward.
1. Based on future action probabilities, so depends on policy
2. Denoted $v_\pi(s) = E[G_t | S_t = s]$ (expectation of reward)

B. The Bellman Equation summarizes all next state values and probabilities to get a state's value

C. To find an optimal policy, π_* one must seek to maximize the expected return or value of the policy ($v_\pi(s) \geq v_{\pi'}(s) \; \forall \; s \in \mathcal{S}$)
aka $v_*(s) = \max_\pi v_\pi(s) \quad \forall \; s \in \mathcal{S}$

D. Similarly, optimal π's share optimal action-value function:
$q_*(s,a) = \max_\pi q_\pi(s,a) \quad \forall \; s \in \mathcal{S}, \; a \in \mathcal{A}(s)$
or $q_*(s,a) = \mathbb{E}[R_{t+1} + \gamma v_*(S_{t+1}) \mid S_t = s, A_t = a]$
when optimal policy is followed after the action

1. Games can be solved in this manner theoretically, but often we have non-Markov problems or simply not enough compute
2. Approximations are often settled for

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